

Square Numbers, Cube Numbers & Roots

WorksheetHub AI
Mathematics · Grade 7

Unit 1 · Number

Easy

Squares, cubes and roots

Name: _____

Date: _____

Score: / 22

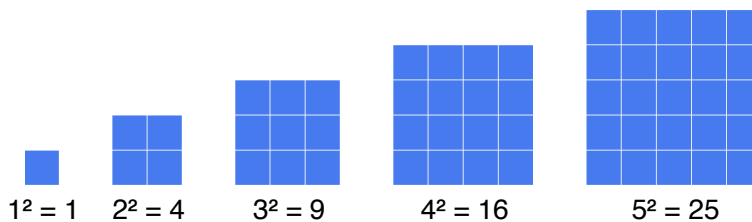


Remember

A **square number** is a number $\times$ itself: $5^2 = 5 \times 5 = 25$.

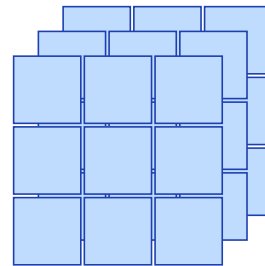
A **cube number** is a number $\times$ itself $\times$ itself: $4^3 = 4 \times 4 \times 4 = 64$.

The **square root** ($\sqrt{\quad}$) undoes squaring: $\sqrt{25} = 5$. The **cube root** ($\sqrt[3]{\quad}$) undoes cubing: $\sqrt[3]{64} = 4$.



Square numbers make a square of dots

n	1	2	3	4	5	6	7	8	9	10
n^2	1	4	9	16	25	36	49	64	81	100
n^3	1	8	27	64	—	—	—	—	—	—



$3^3 = 3 \times 3 \times 3 = 27$ small cubes



Worked example

Find the side of a square with area 49 cm^2 .

Side = $\sqrt{49} = 7 \text{ cm}$ (because $7 \times 7 = 49$).

A. Squares and cubes

1 $6^2 = ?$

Answer:

2 $9^2 = ?$

Answer:

3 $12^2 = ?$

Answer:

4 $15^2 = ?$

Answer:

5 $2^3 = ?$
Answer:

6 $3^3 = ?$
Answer:

7 $5^3 = ?$
Answer:

8 $10^3 = ?$
Answer:

B. Roots

9 $\sqrt{81}$
Answer:

10 $\sqrt{144}$
Answer:

11 $\sqrt{100}$
Answer:

12 $\sqrt[3]{8}$
Answer:

13 $\sqrt[3]{27}$
Answer:

14 $\sqrt[3]{125}$
Answer:

C. Spot the numbers

15 Circle the square numbers: 16, 20, 25, 30, 36, 45, 64
Answer:

16 List all the cube numbers up to 100.
Answer:

17 Fill in: $8^2 = \underline{\quad}$ $\underline{\quad}^2 = 121$ $4^3 = \underline{\quad}$ $\underline{\quad}^3 = 216$
Answer:

D. Real-life problems

18 A square garden has an area of 81 m². How long is each side?
Answer:

19 A cube-shaped box has a volume of 64 cm³. How long is each edge?
Answer:

- 20 A square patio is made of 144 square tiles. How many tiles are along one side?

Answer:

★ Challenge

- 21 Which number under 100 is **both** a square number and a cube number? (Not 1)

Answer:

- 22 Calculate: $3^2 + 4^2$ and compare with 5^2 . What do you notice?

Answer: